

CADzilla

User Reference Manual

S-100 bus graphics controller board — Hitachi HD63484 ACRTC & Brooktree Bt453 video DAC

Document revision 1.2 · 2026-10-01

Preface

CADzilla exists for one reason: to enjoy the graphics board design process the way I did it forty years ago. The circuit architecture, schematic design, and PCB layout were all done by hand, with no AI assistance — the way boards like this were designed back then.

Every component on the board, aside from the modern SRAM used for the frame buffer, is a part that would have been available in 1984, when the Hitachi HD63484 was released and the S-100 bus was still a going concern. As far as I could find, no one had ever built an HD63484-based S-100 board — so I built one.

AI did play a role, but not in the hardware: it made it possible to build a CP/M software development environment and toolchain for the board that I could never have put together on my own.

1. Overview

CADzilla is an S-100 bus board built around the Hitachi HD63484 Advanced CRT Controller (ACRTC) paired with a Brooktree Bt453 video DAC. The board carries 2 MBytes of video frame buffer memory and supports 8-bit-per-pixel (256-color) display modes at resolutions up to 1024×768.

The board occupies a single, jumper/switch-selectable 8-byte block of S-100 I/O space. This manual documents that I/O map, the board's MODE register, the SW1 configuration DIP switch, and the set of HD63484 features the board does and does not support.

2. I/O Address Map

The board decodes a single 8-byte-aligned base I/O address, set by SW1 (see Section 4). All board registers are addressed as offsets from that base address:

Offset	Register	Access
`Base + 0`	HD63484 Address/Status Register (RS = 0)	Read/Write
`Base + 1`	MODE Register (board configuration — see Section 3)	Write only
`Base + 2`	HD63484 Control/FIFO Register (RS = 1)	Read/Write
`Base + 3`	Reserved / not decoded	—
Base + 4 – Base + 7	Bt453 video DAC registers	Read/Write

RS is the HD63484 register-select input: RS = 0 selects the address/status register at Base + 0, RS = 1 selects the control/FIFO register at Base + 2.

3. MODE Register (Base + 1, write only)

The MODE register configures board-level options that are outside the HD63484's own register set. It is write-only.

Bit	Name	Description
0	HSPOL	Horizontal sync polarity: 0 = positive sync pulse, 1 = negative sync pulse
1	VSPOL	Vertical sync polarity: 0 = positive sync pulse, 1 = negative sync pulse
2	AMODE	Address mode: 0 = single access mode, 1 = interleaved access mode
3	OLEN	Overlay enable: 0 = overlay plane blanked (shift register loads zeros), 1 = overlay pixels displayed over the frame buffer
4	OLSEL	Overlay select: 0 = lower half, 1 = upper half of the overlay memory — selects which 64K-word half of the 128K-word overlay SRAM a drawing-mode CHR screen addresses
5–7	—	Reserved, write as 0

The overlay (U19, 1 bit per pixel, mapped 1:1 onto frame-buffer pixels) is drawn by targeting an ACRTC screen with the CHR bit set in its Memory Width register. A CHR screen only has a 64K-word address space, so OLSEL selects which 64K-word half of the 128K-word overlay SRAM that screen addresses. OLSEL is cleared to 0 by reset. A CHR screen must never be displayed — overlay display is driven by the frame buffer's own display cycles, independent of OLSEL.

4. Configuration DIP Switch (SW1)

SW1 is an 8-position DIP switch that sets the board's I/O base address and host interface options.

Switch	Function	On	Off
SW1-1 – SW1-5	I/O base address (address bits 7:3)	—	—
SW1-6	HD63484 host bus width	8-bit host interface: DACK* follows RESET*, so it is low during reset	16-bit host interface: DACK* is pulled high
SW1-7	16-bit address decode enable	Decodes the full 16-bit I/O address (A15–A8 must be 00h)	Decodes only the 8-bit I/O address
SW1-8	Interrupt enable	HD63484 interrupt request connected to the S-100 bus	Interrupt request disconnected from the S-100 bus

SW1-1 through SW1-5 map directly to address bits 7 down to 3 of the board's I/O base address; the low three address bits (2:0) are fixed at zero, so the board's registers always occupy an 8-byte-aligned block (Base + 0 through Base + 7) somewhere in the 256-byte S-100 8-bit I/O space.

5. Frame Buffer

Property	Value
Memory type	2 MBytes of SRAM — no refresh cycles required
Pixel depth	8 bits per pixel only
Graphics Address Increment (GAI)	8
Maximum display resolution	1024 × 768

6. HD63484 Feature Support Summary

CADzilla implements a deliberately reduced subset of the HD63484's capabilities, matched to the board's fixed 8-bit-per-pixel, SRAM-based frame buffer. The table below summarizes what is and is not supported.

Feature	Status
Pixel depth	8 bits per pixel only (GAI = 8)
Frame buffer refresh	Not required (SRAM frame buffer)
Cursors	Not supported
Superimposed (dual-screen) access mode	Not supported
Light pen	Not supported
Zooming	Not supported
Video attributes	Not supported
External sync	Not supported
DMA transfers	Not supported
Interrupts	Supported (enabled via SW1-8, see Section 4)

7. Revision History

Revision	Date	Notes
1.0	2026-09-23	Initial I/O map, MODE register, SW1, and feature support documentation.
1.1	2026-09-24	Added Preface with the board's design philosophy; renamed manual title.
1.2	2026-10-01	Corrected SW1-6: it selects HD63484 host bus width directly (On = 8-bit, DACK* follows RESET*; Off = 16-bit, DACK* pulled high), not a reset-time mode select. Clarified SW1-7's A15-A8 = 00h requirement. Documented MODE register bit 4 (OLSEL), which selects the upper or lower half of the overlay memory for drawing access, and resolved OLEN's prior TBD status.